

Group 9 BEATRIX Project

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Source: John Monahan, 2010

3 Features:

This Project can be broken down into 3 main components.

- Object Tracking
- Audio Detection
- GUI

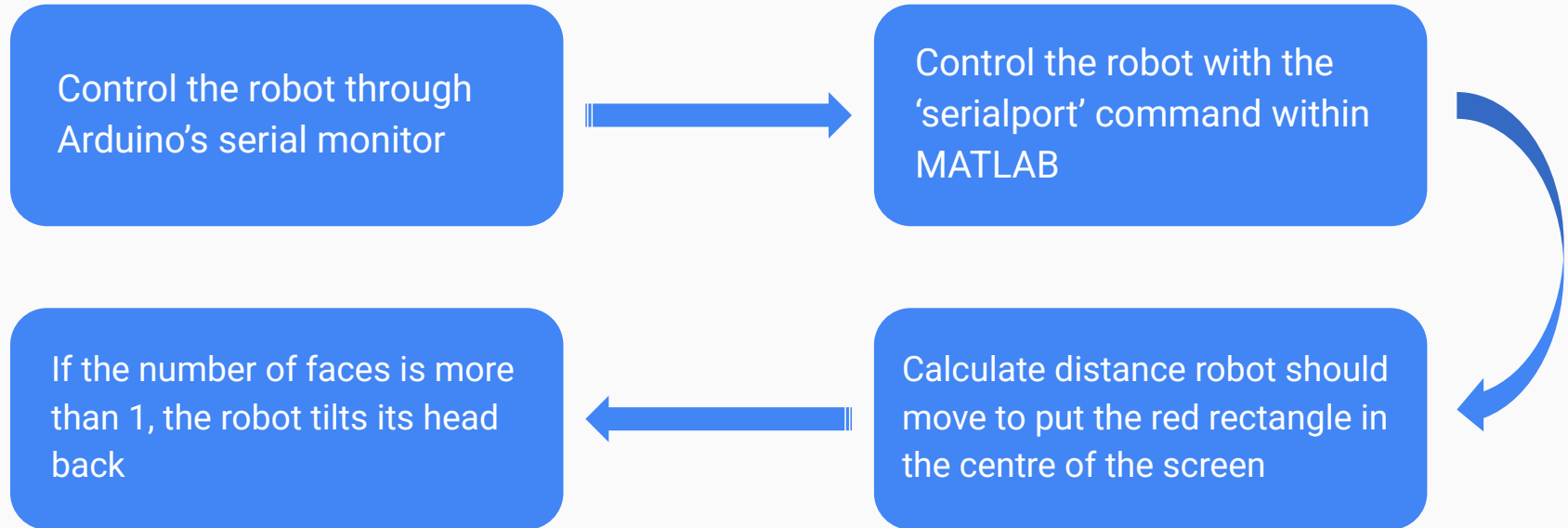
1. Object Tracking Goals:

- Display footage captured by the webcam real time
- Detect faces and highlight them with a red box
- When one person is in frame the robot should react by turning to look at them (vertically or horizontally)
- When multiple people are in frame the robot should react by tilting its head back

Intermediate Task Process:

- Display webcam footage via USB (using the MATLAB support package for USB webcams add-on)
- Using the MTCNN add-on we can detect the locations of faces in frame
- Draw rectangles around any detected faces
- Store the number of faces detected as a variable

Advanced Task Process:



Skills Gained:

- Using a webcam with MATLAB
- Sending information in MATLAB with serialport

Improvements:

- Follow face with a faster response time
- Follows only first face that comes into the frame.

2. Audio Detection Goals:

- Detect sounds using microphones
- Compare amplitude of sound with background noise
- If the sound exceeds a certain threshold, the robot reacts
- The robot dances by rotating left and right



Angry birds 2019

System Process

Home Position Setup

- Manually adjusts the robot to home position

Audio Input

- Microphone sends sound signals to MATLAB

Signal Processing

- Audio samples grouped into processing windows
- Mean Absolute Value (MAV) calculated

Robot Reaction

Beatrix, do a little dance



marve,2022l

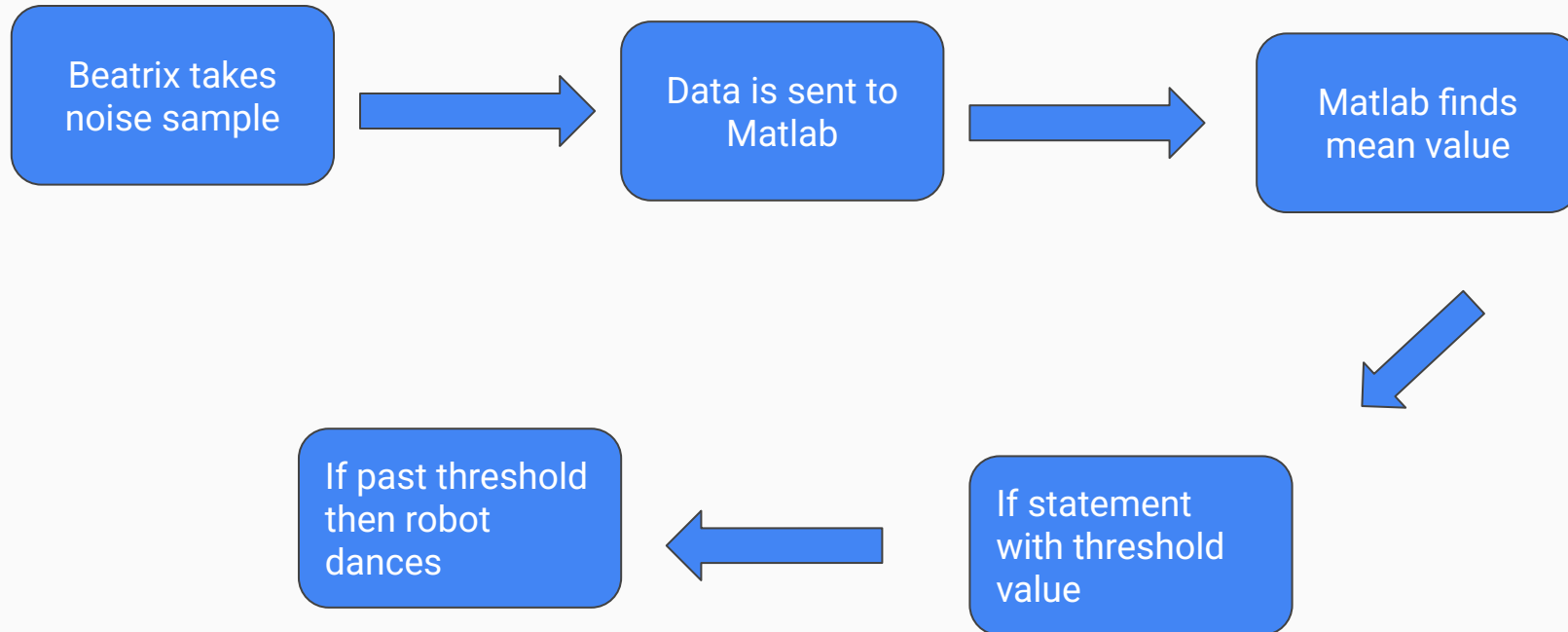
If the MAV is above the threshold:

- Motor rotates left and right using MATLAB playtone() function
- Robot performs dance motion.

Motion:

- Starts from home position
- One of the motors moves a fixed number of steps to the right
- The same motor that moves to the left with double the fixed amount of steps
- The motor then returns to the home position

Robot reaction flow chart



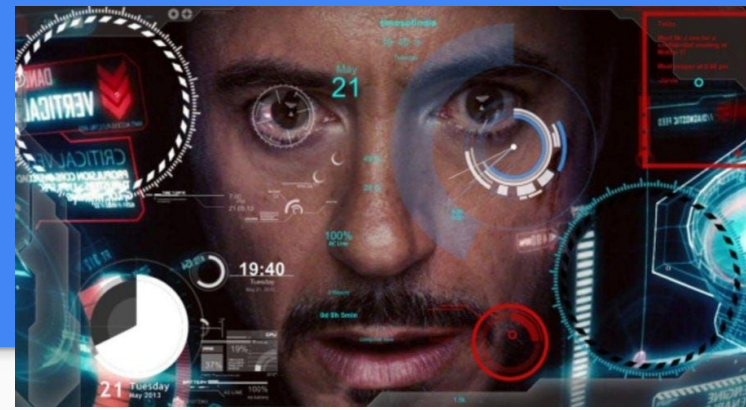
Skills Gained:

- Programming microcontrollers for Matlab
- Risk management skills

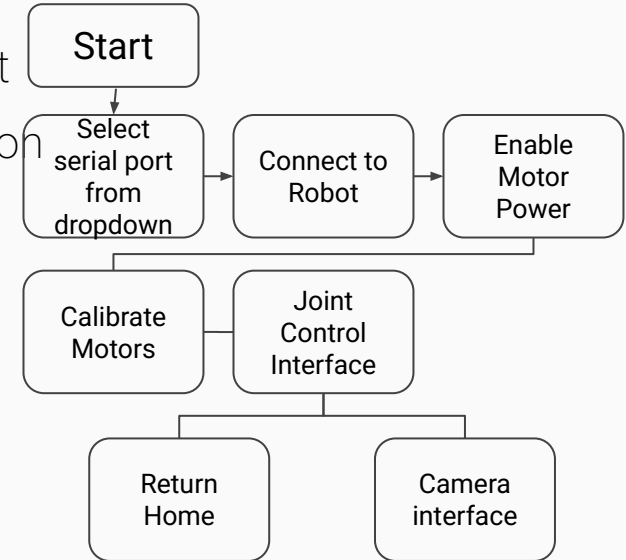
Improvements:

- Make the beatrix follow the direction of sound.
- Adding filter for more accuracy.

3. GUI Goals



- Provide a graphical interface for controlling the robot
- Connect MATLAB to Arduino via serial communication
- Provide manual control of X, Y, Z motors
- Allow users to enable, disable and calibrate motors
- Return the robot to its home position



GUI Implementation



The GUI is divided into four control panels

Arduino Interface

Serial Ports

Cameras Interface

Motors Status

Robot Joint Control

Motor X

Motor Y

Motor Z

Arduino Interface

- Serial port selection
- Connect / Disconnect Arduino

Motor Status

- Motor calibration
- Enable / disable motors

Joint Control

- Manual control of X, Y and Z joints

Camera Interface

- Camera activation
- Image capture
- Edge detection

GUI - Camera Interface

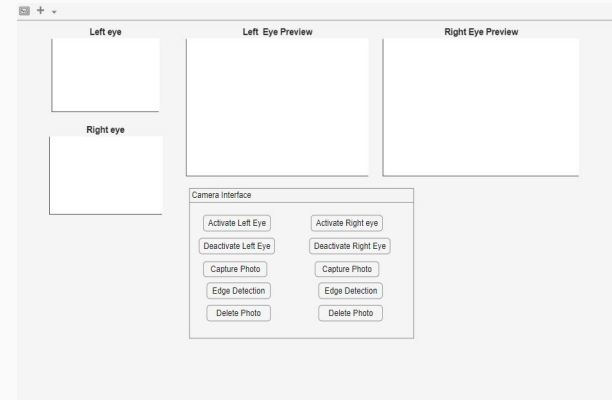
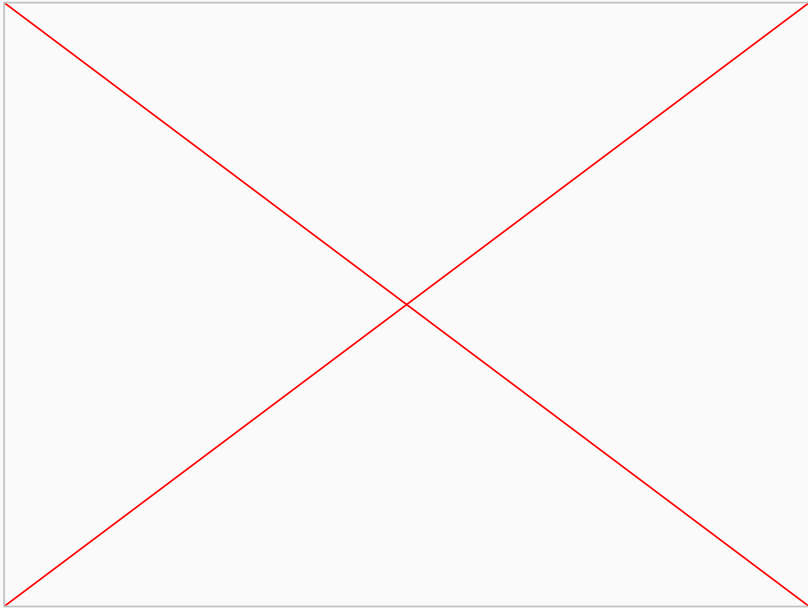
Goal

- Activate a live stream
- Deactivate the live stream and add a background image
- Photo capturing feature
- Edge detection feature
- Delete button

Improvements

- Add a save button for the photos.
- Add a visual indication when a task was successful ie green button when successful and red when failed.
- Use github to share and collaborate the code to have one GUI system. s
- Error message

Demonstration



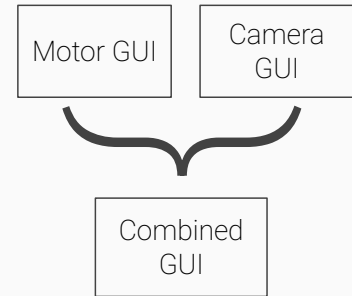
Skills Gained:

- Programming microcontrollers for MATLAB
- Sending and receiving information with Serial port
- Creating interfaces within the MATLAB App designer

Improvements:

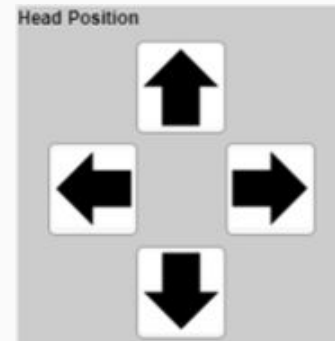
Integrated GUI

- Combine motor and camera interfaces
- Single control environment



Control Pad Navigation

- Directional control pad for manual face positioning
- Faster than entering relative motor values



Thanks for listening!